

SUMMARY TABLE: GENERAL SCIENCE GRADE 10

Sub-Strand 3.3: Waves — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 3.3: Waves
Driving Question	DRIVING QUESTION: How do waves carry energy through our environment — and how can understanding their behaviour protect Kenyan communities, improve technology, and help us make sense of the world around us?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 ANCHORING PHENOMENON AND WAVE VOCABULARY	A narrated scenario of a Kisumu fisherman hearing thunder from behind a cliff and seeing water waves change direction near the shore, illustrated by a photograph of the Kisumu shoreline and a slinky spring demonstration of transverse and longitudinal pulses	The key vocabulary of waves: wavelength, amplitude, period, frequency, wave velocity, transverse wave, and longitudinal wave, each anchored to a labelled diagram and annotated with Kenyan real-world examples including Lake Victoria water waves and spoken Swahili vowel sounds	Learners produced initial (pre-instruction) predictions for the anchoring phenomenon and acknowledged that the new vocabulary gives them a shared scientific language to begin investigating the fisherman puzzle, while recognising that the full explanation (reflection, refraction, and diffraction) will emerge across the unit
Lesson 2 The Wave Equation — Connecting Speed, Frequency and Wavelength	During the slinky demonstration on the classroom floor, learners observed that increasing the rate of hand oscillation (frequency) produced more crests along the spring length — the distance between crests (wavelength) became shorter. The overall wave appeared to travel along the spring at the same speed regardless of how fast the hand moved. In the ripple-tank simulation, learners observed that low-frequency sources produced widely spaced wavefronts while high-frequency sources produced closely spaced wavefronts, with the wave speed remaining constant in the same medium.	Through guided derivation, learners established that in one period T , a wave travels exactly one wavelength λ , giving $\lambda = vT$. Substituting $T = 1/f$ yields the wave equation $v = f\lambda$. This means that in a given medium (constant v), frequency and wavelength are inversely proportional. Learners applied this to four Kenyan contexts: the wavelength of Classic 105 FM (98.4 MHz) radio waves is approximately 3.05 m; the speed of sound at Fort Jesus Mombasa is approximately 340 m/s; a Lake Victoria boat-engine wave ($\lambda = 0.8$ m, $v = 3.2$ m/s) has frequency 4 Hz and period 0.25 s; and Indian Ocean swells ($v = 15$ m/s, $\lambda = 75$ m) have period 5 s, frequency 0.2 Hz, meaning 12 crests per minute reach	The wave equation $v = f\lambda$ explains why the fisherman at Lake Victoria can begin to assign numbers to his observations: the deep-rumble thunder sound has a low frequency and therefore a long wavelength, which — as learners will discover in Lesson 5 — enables it to diffract (bend) around the rocky headland more effectively than high-frequency sounds. The same equation explains why a Nairobi FM radio signal has a wavelength of roughly 3 m, which is relevant to whether that signal can diffract around a Rift Valley hill (a question that will be quantitatively resolved in Lesson 5 on diffraction). Learners cannot yet fully explain the bending mechanism, but they now have the mathematical tool to

		the shore.	characterise any wave in the anchoring phenomenon.
Lesson 3 Reflection of Waves: Why Does Sound Bounce Back — and Could It Explain the Fisherman's Puzzle?	Learners stood at measured distances (20–40 m) from the school perimeter wall, clapped rhythmically, and timed echoes to calculate the speed of sound. Groups pooled their results on a class data board and compared calculated values (expected range: 300–380 m/s) to the accepted value of ~340 m/s.	Sound reflects off hard surfaces; the angle of incidence equals the angle of reflection (law of reflection). Echoes are reflected sound waves. Reflection can explain why the fisherman hears some thunder from the direction of the cliff — but it cannot explain sound reaching the completely shadowed cove behind the headland.	The fisherman's experience of hearing thunder from the direction of the rocky cliff can be partly explained by the reflection of sound waves off the hard rock surface, redirecting some of the sound energy toward him. However, reflection requires a line-of-sight path to the reflecting surface and cannot explain why sound reaches the protected cove with no reflective surface facing the storm — that part of the puzzle remains unsolved and points toward a different wave behaviour (diffraction) to be explored in Lesson 5.
Lesson 4 Refraction of Waves — Why Does the Water Bend at the Shore and the Radio Fade in the Rift Valley?	Learners watched slow-motion ripple-tank footage (Clip A) showing water wavefronts compressing and bending as they crossed from a deep region into a shallow region, and then Mombasa coastal-swell footage (Clip B) showing the same behaviour at the scale of the Indian Ocean. They completed a structured observation table recording changes in wave speed, wavelength, and direction at the boundary, and drew a labelled refraction diagram (incident ray, refracted ray, normal, boundary, wavefronts).	Learners learned that refraction is the change in direction of a wave caused by a change in its speed when it crosses a boundary between two media. The wave bends toward the normal when it slows down and away from the normal when it speeds up. Frequency is conserved at the boundary; wavelength changes in proportion to speed ($v = f\lambda$). A temperature inversion at night causes sound to travel faster in the upper warm layer and refract back downward, explaining why Lamu fishermen hear voices farther across the water at night. Applying $v = f\lambda$: at 680 Hz, a wave travelling at 340 m/s has wavelength 0.50 m; at 360 m/s it has wavelength ≈ 0.529 m.	Refraction explains why the water waves in the Lake Victoria phenomenon change direction as they cross from the deep open lake into the shallow bay near the Kisumu shore — the wave slows down in the shallower water and bends toward the normal at the deep-to-shallow boundary. This is the same mechanism seen in the Mombasa coastal swell. Refraction also partially explains how sound from the distant storm reaches the fisherman at night: a temperature inversion refracts the sound downward, allowing it to travel farther along the water surface. However, refraction alone cannot explain why sound reaches the protected cove behind the solid headland, or why the FM radio signal disappears completely behind a Rift Valley ridge — those require a third wave behaviour (diffraction), which will be the focus of Lesson 5.
Lesson 5 Diffraction of Waves — Sound Around the Corner and Signals Over the Hill	Learners carried out a classroom doorway experiment in which a speaker said 'Habari za asubuhi!' just outside an open doorway while listeners positioned around the corner (out of line of sight) rated audibility at three gap widths. They also watched and sketched a water-wave diffraction simulation showing wavefronts spreading through gaps of decreasing width.	Learners constructed wavefront diagrams showing that waves spread (diffract) around obstacles and through gaps, and that diffraction is greatest when the gap width is close to the wavelength. They connected diffraction to why sound reaches the Kisumu fisherman around the rocky headland and why mobile/FM radio signals can be detected behind hills in the Rift	Learners updated the class Driving Question Board with an integrated three-property explanation of the anchoring phenomenon (reflection = echo off cliff; refraction = water-wave bending at shallow bay; diffraction = sound spreading around headland into cove) and added a green-coded diffraction layer to their cumulative Kisumu bay wavefront diagram.

		Valley.	
Lesson 6 Applications of Wave Properties in Real Life	In their expert groups, learners researched and presented one of four real Kenyan wave applications: KMFRI SONAR depth-sounding on Lake Victoria; ultrasound scanning at Kenyatta National Hospital; cats-eye reflectors and speed cameras on the Nairobi–Mombasa highway; and seismic wave monitoring by the Kenya Meteorological Department. They created posters or digital slides including wave diagrams, property labels, Kenyan context details, and a connection sentence to the Lake Victoria fisherman phenomenon, then presented to the class and responded to peer questions.	Each real-life Kenyan application deliberately exploits one or more wave properties. SONAR and ultrasound use REFLECTION — a directed pulse is sent out and the returning echo carries information about distance or density. Cats-eye reflectors and speed cameras use REFLECTION of light waves from a curved surface back toward its source. Seismic wave monitoring uses REFRACTION (waves bend as they pass through rock layers of different density) and DIFFRACTION (waves spread around geological structures) — the same two properties that explain why the fisherman's water waves entered the sheltered cove and why sound bent around the rocky headland. All applications are grounded in the same wave equation and the same three properties investigated in Lessons 3–5.	Wave properties are not abstract laboratory phenomena — they are the physical mechanisms inside medical imaging machines, marine research vessels, road safety infrastructure, and earthquake detection systems across Kenya. The fisherman on Lake Victoria was not experiencing something unusual; he was experiencing the same reflection, refraction, and diffraction that Kenyan engineers and scientists harness every day. Understanding these properties at a mechanistic level — knowing WHY a wave bends, bounces, or spreads — is what allows technology to be designed, improved, and used safely for the benefit of Kenyan communities.
Lesson 7 Effects of Waves on the Environment & Mitigation	Learners researched print and non-print sources to find evidence of wave impacts on the Kenyan environment: coastal erosion at Mombasa and Lamu driven by wave diffraction and high-amplitude storm waves; seismic wave damage to Nairobi infrastructure linked to wave frequency and building resonance; noise pollution from low-frequency sound waves near Kisumu port and along the Thika Superhighway; and disruption of Lake Victoria's tilapia and Nile perch spawning grounds caused by wave refraction in shallow water and boat-wake wave energy.	Each environmental impact is caused by a specific, nameable wave property: diffraction concentrates wave energy on unprotected shorelines; high-amplitude waves carry sufficient energy to erode coral-stone foundations; seismic wave frequency can match structural resonance frequencies in buildings; noise is a longitudinal wave whose energy disrupts biological systems. Mitigation measures work by applying the same wave physics in reverse — sea walls reflect or absorb wave energy before it reaches the shoreline; building codes specify materials whose resonance frequencies differ from local seismic wave frequencies; noise barriers reflect and absorb sound waves; marine protected areas reduce boat-wake wave disturbance of shallow spawning grounds.	Learners used the wave equation and wave-property vocabulary from Lessons 1–6 to construct causal arguments in a structured debate about the environmental trade-offs of a Kenyan coastal wave-energy power station. The debate required learners to weigh the same wave energy that causes coastal erosion as a potential renewable electricity source — illustrating that wave energy is neither inherently harmful nor beneficial, but that human decisions about technology and mitigation determine outcomes. The fisherman phenomenon was extended: the same diffraction and refraction processes that made the storm night puzzling are the processes that, sustained over decades, reshape Kenya's coastlines, threaten Lamu's UNESCO heritage buildings, and alter the ecology of Lake Victoria. Learners updated the DQB with environmental-impact and mitigation cards and revised their cumulative models to include a human-mitigation layer, advancing the class's collective explanation

			of the driving question.
Lesson 8 Model Building, Numerical Consolidation & DQB Closure — How Wave Science Keeps Kenyan Communities Safer	In this unit, students observed: sound reflecting off the rocky headland at Lake Victoria (echo experiment), water waves refracting as they crossed from deep open lake into the shallow Kisumu bay (wavefront diagrams and simulation), sound waves diffracting around a corner (classroom experiment with a sound source behind a barrier), and the fading of mobile-phone signals in Rift Valley valleys behind hills (case study and discussion).	Students learned: the wave equation $v = f\lambda$ and how to use it to calculate speed, frequency, and wavelength in Kenyan contexts (Nairobi FM radio, Mombasa sonar, Kisumu water waves); the law of reflection and the echo-distance formula $d = (v \times t) / 2$; that refraction changes wave speed and direction but NOT frequency, and that wavelength shifts proportionally with speed; that diffraction spreads waves around obstacles and through gaps, and is greatest when wavelength $\approx$ gap size; that waves carry energy and can damage coastal communities (storm surges, coastal erosion), but that this impact can be mitigated through mangrove restoration, wave-energy barriers, and early-warning sonar systems.	Students can now explain the full anchoring phenomenon: the Kisumu fisherman hears thunder from behind the cliff because sound waves DIFFRACT around the rocky headland — the low-frequency rumble has a long wavelength comparable to the headland gap, so it spreads into the cove. The water waves change direction at the edge of the shallow bay because of REFRACTION — the wave speed decreases in shallow water, bending the wavefronts toward the normal, steering wave energy into the protected cove even without a straight-line path from the storm. The mobile-phone signal that fades behind a Rift Valley hill behaves the same way as the sound around the headland — microwaves DIFFRACT around the hill, though imperfectly, which is why engineers place repeater towers on hilltops to boost the diffracted signal into valleys. Wave science therefore directly answers the driving question: understanding reflection, refraction, and diffraction — and the mathematics linking them through $v = f\lambda$ — equips communities, engineers, and scientists to protect lives, design better communications, and make sense of the energy-filled world around them.

ATTRIBUTION AND LICENSING

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